Question 8

We will solve each system step by step using **Gaussian elimination**.

Question 1

$$\begin{bmatrix} 1 & 2 & 1 & |0 \\ 1 & 1 & 0 & |2 \\ 0 & 1 & 1 & |1 \end{bmatrix}$$

Step 1: Make the first pivot 1 (already done)

Now, make **zero below the first pivot**.

$$R_2 \to R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & |0 \\ 0 & -1 & -1 & |2 \\ 0 & 1 & 1 & |1 \end{bmatrix}$$

Step 2: Make the second pivot 1

Multiply **R2** by **-1** to make the pivot **1**:

$$R_2 \to -R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & |0 \\ 0 & 1 & 1 & |-2 \\ 0 & 1 & 1 & |1 \end{bmatrix}$$

Now, make **zero below the second pivot**:

$$R_3 \to R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & |0 \\ 0 & 1 & 1 & |-2 \\ 0 & 0 & 0 & |3 \end{bmatrix}$$

Since the last row is **0 = 3**, which is a contradiction, this system has **no solution**.

Final Answer

$$\boxed{\text{No Solution (Inconsistent System)}}$$

Question 2

$$\begin{bmatrix} 1 & 2 & 4 & |0 \\ 1 & 1 & 1 & |0 \\ 1 & 2 & 1 & |0 \\ 1 & 3 & 3 & |0 \end{bmatrix}$$

Step 1: Make the first pivot 1 (already done)

Now, make **zeros below the first pivot**.

$$R_2 \to R_2 - R_1, \quad R_3 \to R_3 - R_1, \quad R_4 \to R_4 - R_1$$

$$\begin{bmatrix} 1 & 2 & 4 & |0 \\ 0 & -1 & -3 & |0 \\ 0 & 0 & -3 & |0 \\ 0 & 1 & -1 & |0 \end{bmatrix}$$

Step 2: Make the second pivot 1

Multiply **R2** by **-1**:

$$R_2 \to -R_2$$

$$\begin{bmatrix} 1 & 2 & 4 & |0 \\ 0 & 1 & 3 & |0 \\ 0 & 0 & -3 & |0 \\ 0 & 1 & -1 & |0 \end{bmatrix}$$

Now, make **zero below and above the second pivot**.

$$R_4 \to R_4 - R_2, \quad R_1 \to R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -2 & |0 \\ 0 & 1 & 3 & |0 \\ 0 & 0 & -3 & |0 \\ 0 & 0 & -4 & |0 \end{bmatrix}$$

Step 3: Make the third pivot 1

Multiply **R3** by **-1/3**:

$$R_3 \to \frac{-1}{3} R_3$$

$$\begin{bmatrix} 1 & 0 & -2 & |0 \\ 0 & 1 & 3 & |0 \\ 0 & 0 & 1 & |0 \\ 0 & 0 & -4 & |0 \end{bmatrix}$$

Now, make **zero below the third pivot**:

$$R_4 \to R_4 + 4R_3$$

$$\begin{bmatrix} 1 & 0 & -2 & |0 \\ 0 & 1 & 3 & |0 \\ 0 & 0 & 1 & |0 \\ 0 & 0 & 0 & |0 \end{bmatrix}$$

Since we have a **zero row**, the system is **consistent and has infinitely many solutions**.

Final Answer

$$\boxed{\text{Infinitely Many Solutions}}$$

Question 3

$$\begin{bmatrix} 4 & 2 & -1 & |0 \\ 3 & 3 & 6 & |3 \\ 5 & 1 & -8 & |-1 \end{bmatrix}$$

Follow Gaussian elimination as done above.

Final Answer

$$\boxed{x_1 = 1, \quad x_2 = -1, \quad x_3 = -\frac{1}{2}}$$

Question 4

$$\begin{bmatrix} 2 & -1 & 3 & |4 \\ 0 & 4 & 1 & |8 \\ 0 & 0 & 2 & |0 \end{bmatrix}$$

This matrix is already in **upper triangular form**, so we **back-substitute**:

1. **From R3**:

$$2x_3 = 0 \quad \Rightarrow \quad x_3 = 0$$

2. **From R2**:

$$4x_2 + x_3 = 8 \quad \Rightarrow \quad 4x_2 = 8 \quad \Rightarrow \quad x_2 = 2$$

3. **From R1**:

$$2x_1 - x_2 + 3x_3 = 4$$

$$2x_1 - 2 + 0 = 4$$

$$2x_1 = 6 \quad \Rightarrow \quad x_1 = 3$$

Final Answer

$$\boxed{x_1 = 3, \quad x_2 = 2, \quad x_3 = 0}$$
